

Guodong DU

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SUMMARY

PhD researcher with **11 first / co-first / co-corresponding author papers** at ACL, NeurIPS, ICLR and EMNLP. Expertise in **LLM post-training & alignment**, **model merging & knowledge fusion**, and **training & inference efficiency**; increasingly focused on **LLM agents** — agentic & multi-turn RL, long-horizon planning, and **memory systems** (spanning code/tool agents and embodied AI). Industry experience at Tencent Youtu, Intellifusion, TeleAI, and Huawei.

RESEARCH INTEREST

- **LLM & Code Agents:** agentic & multi-turn RL (RLVR / GRPO), executable environments & trajectory data, tool use.
- **Agent Memory & Long-horizon Planning:** memory-augmented planning, skill consolidation, fast-slow systems.
- **Model Merging, Knowledge Fusion & Efficiency:** combining, compressing and transferring capabilities across LLMs / MLLMs.

EDUCATION

The Hong Kong Polytechnic University (PolyU) PhD student (Computing)	Hong Kong, China <i>Aug.2025 - Jun.2029</i>
National University of Singapore (NUS) M.Sc. (Electrical & Computer Engineering)	Singapore <i>Aug.2018 - Jun.2019</i>
University of Electronic Science and Technology of China (UESTC) B.Eng., GPA: 3.91/4.0, Rank: 12/550 (top 3%)	Chengdu, China <i>Aug.2014 - Jun.2018</i>
City University of Hong Kong (CityU) Non-degree Undergraduate Exchange, GPA: 4.0/5.0	Hong Kong, China <i>Aug.2016 - Jan.2017</i>

SELECTED PUBLICATIONS

(* equal contribution † corresponding)

NestedVLA: Learning to Consolidate and Generate Skills for Vision-Language-Action Models.

Guodong Du, Wanyu Lin[†].

NeurIPS 2026

Dynamic Model Merging Made Slim (DiDi-Merging).

Guodong Du, Wanyu Lin[†].

NeurIPS 2026

Knowledge Fusion of Large Language Models via Modular SkillPacks.

Guodong Du, Zhuo Li, ..., Jing Li[†]

ICLR 2026

Parameter Competition Balancing for Model Merging (PCB-Merging).

Guodong Du, Junlin Lee, Jing Li[†], ..., Ho-Kin Tang[†]

NeurIPS 2024

Neural Parameter Search for Slimmer Fine-Tuned Models and Better Transfer.

Guodong Du, ..., Jing Li[†]

ACL 2025

Knowledge Fusion by Evolving Weights of Language Models.

Guodong Du, ..., Sim Kuan Goh[†], Ho-Kin Tang[†]

ACL 2024 Findings

Multi-objective Large Language Model Alignment with Hierarchical Experts (HoE).

Zhuo Li, Guodong Du^{*}, ..., Jing Li[†]

ICLR 2026

Multi-Modality Expansion and Retention for LLMs through Parameter Merging and Decoupling.

Junlin Lee, Guodong Du^{*}, ..., Jing Li[†]

ACL 2025

Skill Weaving: Efficient LLM Improvement via Modular Skillpacks.

Zhuo Li^{*}, Guodong Du^{*}, ..., Jing Li[†]

ACL 2026 Findings

To See a World in a Spark of Neuron: Disentangling Multi-Task Interference for Training-Free Model Merging.

Zitao Fang, Guodong Du[†], ..., Sim Kuan Goh[†]

EMNLP 2025

D-QReLO: Training- and Data-Free Delta Compression for Large Language Models via Quantization and Residual Low-Rank Approximation.

Junlin Lee, Shuangyong Song, Guodong Du[†], ..., Xuelong Li[†]

ACL 2026 Findings

RESEARCH EXPERIENCE

The Hong Kong Polytechnic University (PolyU)

PhD Student

Aug.2025 - Present

- LLM post-training & alignment; multimodal LLMs and agents.
- Embodied intelligence / Vision-Language-Action (current direction).

Knowledge and Language Computing Lab, HITSZ

Advisor: Prof. Jing Li

Research Assistant

Dec.2023 - Aug.2025

- Model merging, knowledge fusion, and preference / alignment learning for LLMs.
- Multimodal LLM knowledge fusion and decoupling; parameter-efficient training.

Learning and Vision Lab, ECE, NUS

Advisor: Prof. Jiashi Feng / Xinchao Wang

Master / Research

Aug.2018 - Sep.2021

- Efficient and lightweight model architectures; low-level computer vision.

INDUSTRY EXPERIENCE

Tencent Youtu Lab

Multimodal Code Agents

Research Intern

Aug.2026 - Sep.2026

- Built and curated datasets for **WorkBuddy**, a multimodal code agent for office tasks.
- Improved **multimodal evaluation** and contributed to **agent-loop assessment**, combining automated scoring with human assessment.

Intellifusion

Research Intern

Embodied AI / VLA

Nov.2025 - Present

- **Long-horizon household manipulation** (BEHAVIOR-1K benchmark): decomposed long-horizon tasks into distinct skills (navigation, manipulation, ...) to drive hierarchical planning.
- Trained **chain-of-thought (CoT)** reasoning for the $\pi_{0.5}$ **VLA model**, and applied **reinforcement learning** to refine fine-grained manipulation.
- Designed a **fast-slow (dual-system)** architecture and improved the planner's **memory system**, boosting end-to-end task success rate.

TeleAI, China Telecom

Research Intern — LLM Post-training

Mentor: Prof. Jing Li

Sep.2024 - May.2025

- **Anti-fraud multi-turn dialogue system** (deployed for trial): an LLM runs multi-turn voice call-backs to suspected scam victims, replacing human agents.
- Built a **dialogue quality-inspection Agent** — an agentic evaluation pipeline with a reasoning reward model for multi-dimensional, interpretable scoring of multi-turn responses.
- Built **multi-turn RL** training on **veRL** (PPO / GRPO) with the user simulator and reward model, raising the **3B model's task-completion rate to 88%**.

Earlier industry R&D (2018–2021): Huawei Noah's Ark Lab & Singapore (image/video SR, ISP); BioMind (medical imaging); YoueData (recommendation).

AWARDS

- 2021, **3rd Place**, NTIRE Video Super-Resolution Challenge (CVPR 2021 Workshop).
- 2017 / 2016, National Encouragement Scholarship; 2015, National Scholarship.
- 2016, Honorable Mention, Mathematical Contest in Modeling (MCM).

ACADEMIC SERVICE

Reviewer: ICLR 2024–2026, NeurIPS 2023–2025, ICML 2025–2026, ACL 2024–2025, EMNLP, etc.
TA: EE5907 Pattern Recognition (NUS ECE).